

MedAssist

Executive Summary

GUIDING PATIENTS FROM HOME, TO THE TREATMENT ROOM

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1. EXECUTIVE OVERVIEW

MedAssist is a web-based patient journey companion (Progressive Web App) that guides each patient through every stage of their hospital visit - from the moment an appointment is confirmed to the conclusion of treatment. The system is delivered through a personalized SMS link: one tap opens a tailored, browser-native experience requiring no app download, no account registration, and no password.

The platform operates across two complementary tracks:

- **Elective track:** Patients with a scheduled appointment receive a Magic Link days before their visit - timed to the first preparation requirement of their specific.
The full flow covers: preparation checklist, digital pre-admission forms, landmark-based navigation, and a live waiting-room status screen.
- **Emergency track (ER):** For walk-in patients, staff generate an instant Magic Link from the back-office panel. The link, valid for 12 hours, opens directly to a waiting screen with real-time status and immediate instructions.

Both tracks share a single back-office dashboard that gives clinical staff a real-time view of all waiting patients, one-tap status updates, and the ability to broadcast messages to all patients in a department simultaneously.

2. THE PROBLEM, TARGET AUDIENCE & DEMAND VALIDATION

The Problem

Medical technology advances every year, yet the patient experience inside hospitals remains structurally unchanged. Three interlocking failures occur every time a patient walks through the door:

\$150B Annual cost of patient no-shows in the US (Artera, 2026)	4,500 Staff hours per hospital per year spent solely on patient wayfinding (Emory, 2025)	\$556K Average annual revenue loss per hospital from cancellations and poor navigation (Signes, 2026)	60–92% of patients experience moderate-to-severe anxiety before a medical procedure
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- **"The Black Hole":** Sitting outside a treatment room with zero information about queue status causes acute uncertainty, frustration, and in documented cases- patients leaving before receiving care.
- **Cognitive overload on arrival:** Identical corridors, confusing signage, and the complete absence of personalized guidance cause patients to get lost precisely when their cognitive capacity is at its lowest due to stress.

- **Unprepared arrivals:** A patient who was unaware of fasting requirements does not only delay themselves, they disrupt the entire day's patient flow and place additional pressure on already-stretched clinical staff.

Target Audience

Primary (B2B): General hospitals, ambulatory clinics, and private medical centers seeking to improve patient experience metrics, reduce no-show and cancellation rates, and redirect clinical staff time from wayfinding to actual care.

Secondary (end-user): Patients aged 18–80 with a scheduled elective procedure, and ER walk-in patients. The platform was designed specifically to work for users with limited digital literacy- no account, no password, RTL Hebrew support, landmark-photo navigation.

Demand Validation

In-depth qualitative interviews were conducted with two senior healthcare practitioners:

Interviewee	Role	Key Finding
Dr. Ofer Hendin	Head of Innovation, Barzilai Medical Center	"Internal navigation is one of the biggest challenges for both patients and staff today - and it's a real gap in the field."
Shira Segev	Radiology Shift Supervisor, Sheba Tel HaShomer	"The hospital would want to adopt this. There's no existing system that combines all these functions together in an accessible way."

Both interviewees confirmed strong appetite for the product's core features. The primary adoption barrier identified was integration with government health IT systems, a challenge MedAssist addresses by operating in parallel to existing systems without requiring direct integration at the initial rollout stage.

3. THE PRODUCT & VALUE PROPOSITION

MedAssist is built around a single design principle: the system reaches out to the patient at the right moment, it does not wait for the patient to look for it. Every module is triggered at the specific point in the patient journey where uncertainty or cognitive load peaks.

Module	What the Patient Receives	Value to the Hospital
Magic Link Entry	Personalized greeting: name, department, date & time. No login required.	Zero adoption friction. Maximum uptake from day one.
Preparation Checklist	Procedure-specific list: what to bring, fasting instructions, medications to pause. Urgent items flagged.	Fewer cancellations due to unpreparedness. Patients arrive ready.
Digital Pre-Admission Forms	Fill questionnaires, provide e-signature, photograph ID and insurance card — before arriving.	Paperwork completed in advance. Faster check-in. Reduced admin load.

Landmark-Photo Navigation	Real photos of waypoints along the route to the department. One image + one instruction per screen.	Significant reduction in staff time spent on patient directions.
Live Waiting Screen	Queue position, estimated wait time (auto-refreshes every 60 s), team broadcasts, pre-set messaging buttons.	Lower anxiety. Fewer "how much longer?" interruptions to clinical staff.
Staff Back-Office	Real-time patient list: name, phase, wait time. One-tap status updates. Department-wide broadcast.	Full operational visibility. Simple queue management. No extra hardware.

The system enforces a maximum of 3 taps per primary action and displays no more than 4 SMS notifications per visit, both constraints that emerged directly from behavioral research into patient stress responses.

4. INNOVATION & KEY DIFFERENTIATORS

Zero-friction access (Magic Link + PWA): A patient under stress will not download an app or remember a password. An SMS link that opens a browser-native experience is the only access model that reliably works under hospital conditions. No app store, no account, no barrier.

Landmark-photo navigation (not GPS): Indoor GPS is unreliable, hardware-dependent, and frustrating when it fails. Humans navigate by visual landmarks. MedAssist uses real photos of physical waypoints, the same way one human gives another directions. This also eliminates accessibility barriers related to age, language, or digital literacy.

Zero-search information architecture: If a patient needs to search for information, the design has failed. Every screen presents exactly the right content for the right moment in the journey, no menus, no navigation trees.

Offline-first PWA: Dead zones inside large hospital buildings are common. The checklist and navigation modules are cached via a Workbox service worker and function fully without an internet connection.

Dual-sided platform (patient + staff): Unlike solutions that address only the patient side, MedAssist includes a full clinical back-office. Every patient action is reflected in the staff dashboard in real time, creating a closed feedback loop between the two sides.

Behavioral science-driven design: Two psychology graduates are embedded in the product team. Every UX decision, from the 3-tap maximum to the phrasing of the waiting screen, was evaluated through the lens of behavioral science and stress psychology.

5. COMPETITIVE LANDSCAPE & ALTERNATIVE SOLUTIONS

No existing solution covers the complete patient journey end-to-end without requiring installation, registration, or dedicated hardware. The table below maps the key alternatives and their gaps:

Solution	Category	Critical Gap vs. MedAssist
HIS / EMR systems (Hadassah, Sheba internal portals)	Existing hospital IT	Staff-facing only. No patient wayfinding, no journey guidance, no live waiting status.
Indoor wayfinding apps (Phunware, Pointr)	Navigation	Requires app installation and expensive Beacon/BLE hardware. No preparation flow, no waiting module.

Patient portals (MyChart, Clalit Online)	Health information	Requires account creation and password. No navigation. No live queue. No real-time visit guidance.
SMS reminder services (Artera, Klara)	Appointment reminders	Passive notification only. No interactive guidance, no navigation, no waiting screen.
Current reality: staff + signage	Status quo	4,500+ staff hours/year. Confusing signage. Zero confirmation that patients are in the right place.

MedAssist's competitive moat lies in the combination: zero-friction access + full journey coverage + no hardware dependency + dual-sided platform - delivered as a single product.

6. TARGET MARKET & MARKET POTENTIAL

\$32B Global Patient Experience Technology market, 2024 (Grand View Research)	17% Projected CAGR through 2030	~80 General hospitals in Israel (primary domestic market)	\$1M+ Estimated ARR potential from 10 hospitals alone
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- Initial target (Israel):** General hospitals and ambulatory clinics in Israel. Preferred entry points: high-throughput departments with significant patient flow, such as: orthopedics, radiology, day surgery. These departments have the clearest pain point (volume × complexity × time pressure) and the most measurable improvement potential.
- Secondary target (international):** Health systems facing the same structural challenges in the US, Western Europe, and Australia. The platform's no-integration-required design means geographic expansion does not depend on local EMR compatibility.

7. GO-TO-MARKET STRATEGY

The strategy follows a land-and-expand model designed to generate real-world evidence before scaling:

Phase	What	Success Metric
1. Pilot	Deploy in one department at a partner hospital. Staff onboarding under 2 hours.	Patient satisfaction score, reduction in staff wayfinding interruptions, % of patients arriving prepared.
2. Expand	Roll out to all departments in the same hospital.	Hospital-level case study with quantified ROI (staff hours saved, cancellation rate reduction).
3. Land & Expand	Annual B2B SaaS contracts. Pricing: per department or per active patient/month.	Signed contracts with 3-5 hospitals. ARR baseline established.
4. Channel	Distribution via HMOs (Clalit, Maccabi) and private hospital networks.	Geographic scale without proportional increase in direct sales cost.

Key go-to-market insight: The biggest adoption barrier identified in our interviews is not price or functionality, it is the fear of complex EMR integration. MedAssist sidesteps this entirely: staff update patient status manually in the

back-office, and the system runs in parallel to existing infrastructure. There is nothing to "integrate" in the first deployment.

8. TEAM & RESPONSIBILITIES

The team combines a strong technical engine with embedded behavioral science expertise, an uncommon combination in the HealthTech space:

Name	Role	Background
Roy Davidovitch	Project Management	B.Sc. Computer Science
Tal Ayoun	Full-Stack Development	B.Sc. Computer Science
Kim Neishtut	Behavioral Research & UX Design	B.A. Psychology
Yasmin Dionina	Content Strategy & Support	B.A. Psychology

Academic Mentor: Dror Margalit, Tel Aviv-Yafo Academic College

The presence of two psychology graduates in a product team is deliberate and central to the product's design philosophy. Every interaction model in MedAssist, from the maximum-3-taps rule to the exact wording of the waiting screen, was reviewed through the lens of behavioral science and stress psychology.

This is not a feature, it is the architecture.